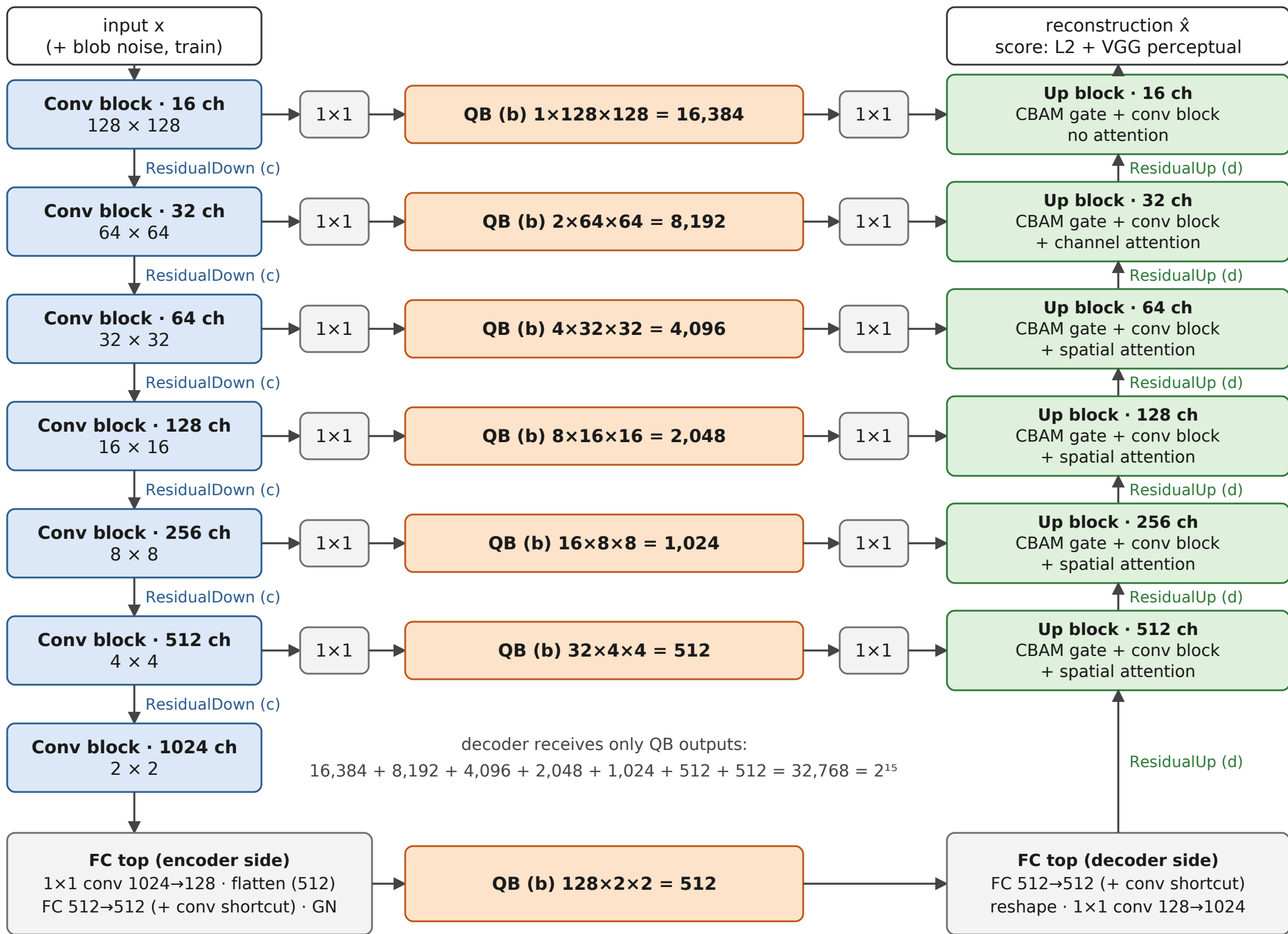
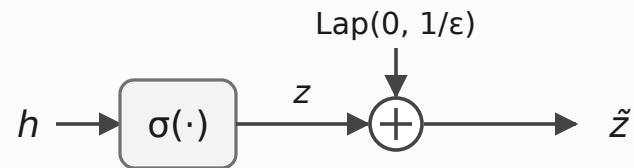


(a) Overall architecture (7 levels, 128 × 128 input)

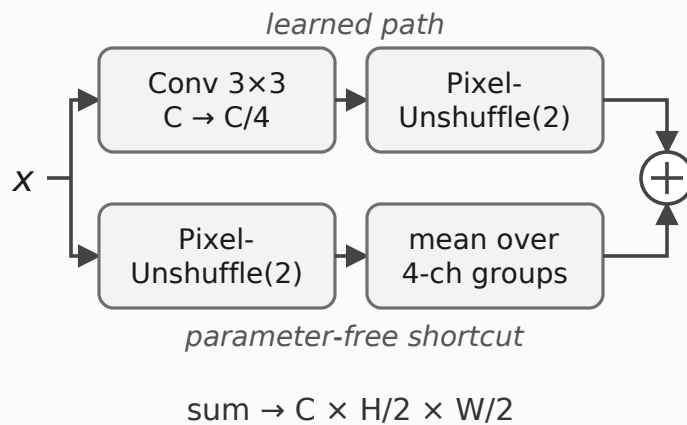


(b) QB layer



train: $\tilde{z} = \sigma(h) + n$, $n \sim \text{Lap}(0, 1/\epsilon)$
test: same (noise kept, 8 draws averaged)
 ϵ -local DP per channel ($\Delta = 1$);
 $I(X; \tilde{X}) \leq N \cdot c(\epsilon)$

(c) ResidualDown $C \times H \times W \rightarrow C \times H/2 \times W/2$



(d) ResidualUp $C \times H \times W \rightarrow C' \times 2H \times 2W$

